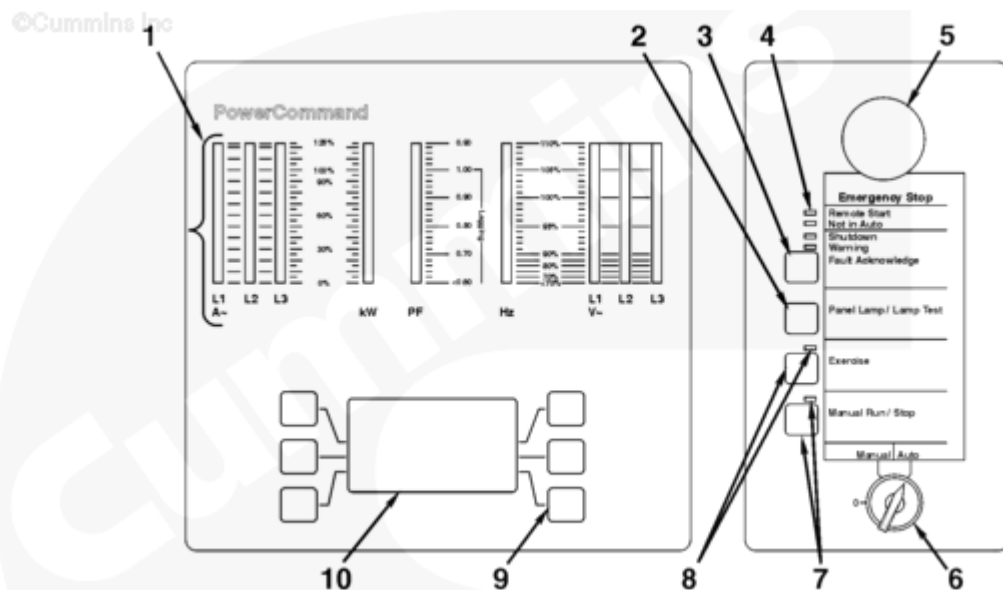


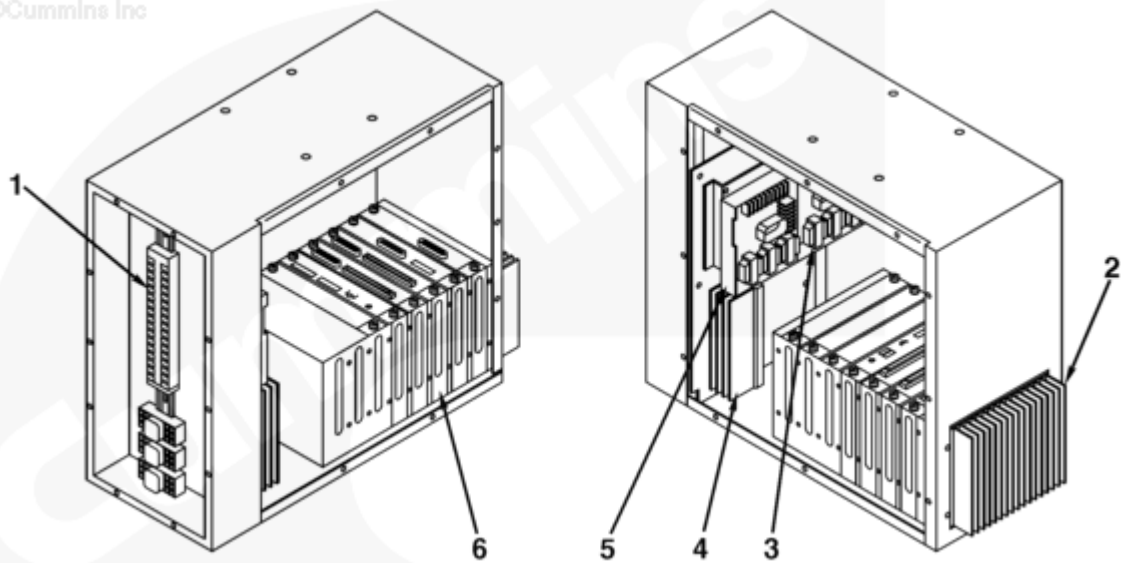
## Trainings ▾

### Generator Views



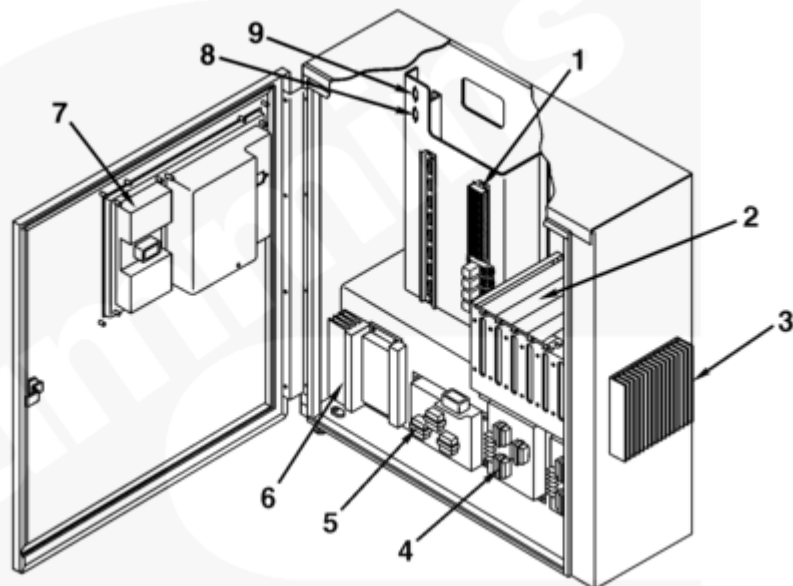
Operator Interface Panel

1. Digital VAC metering panel
2. Panel Lamp/Lamp Test button
3. Fault Acknowledge button
4. Remote Start, **Not** in Auto, Shutdown, Warning status
5. Emergency Stop push button
6. 0/Manual/Auto switch
7. Manual Run/Stop button indicator
8. Exercise button indicator
9. Display menu selection buttons (1 of 6)
10. Graphical display.



Control Housing 1

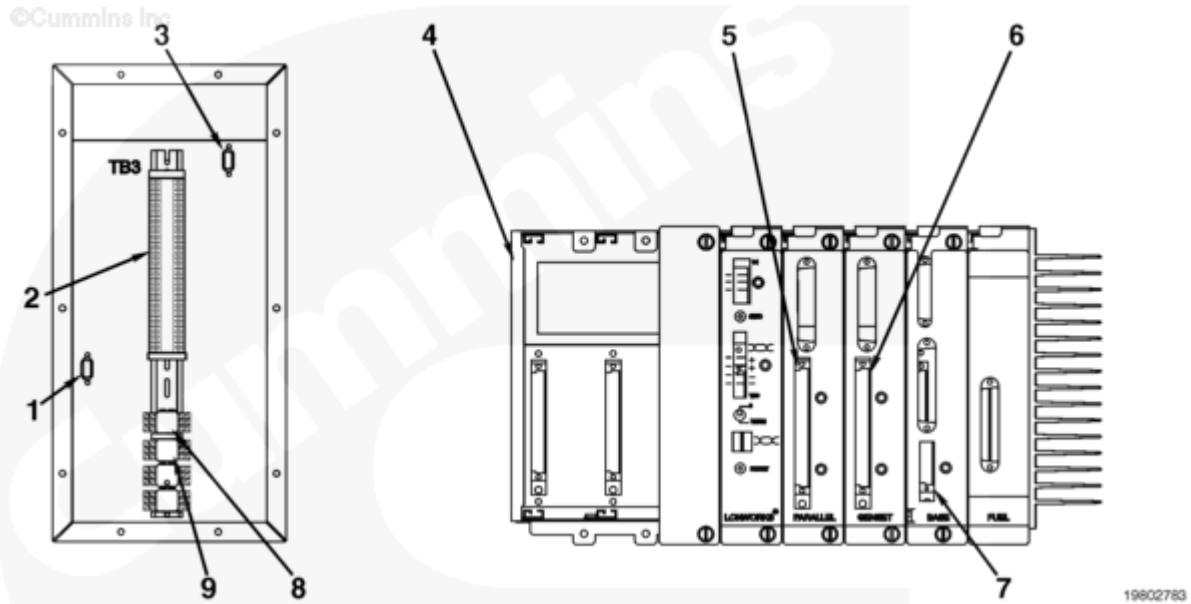
1. TB3
2. Fuel board heat sink
3. PT/CT board
4. Voltage regulator output module
5. Bus PT module
6. Card cage.



Control Housing 2

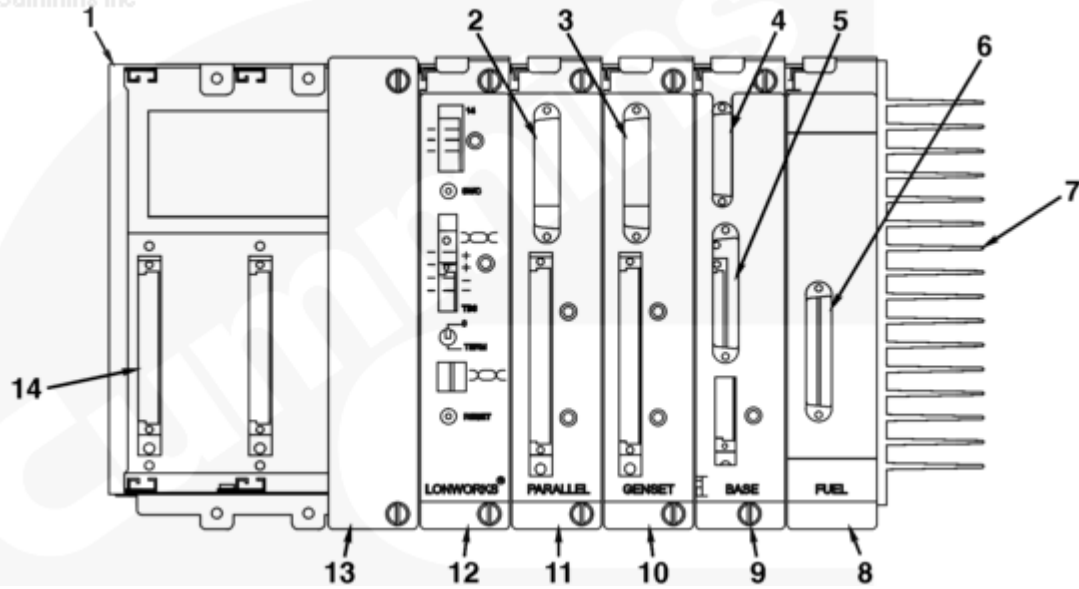
1. TB3
2. Card cage
3. Heat sink
4. Bus PT module
5. PT/CT board

- 6. Voltage regulator
- 7. Operator interface panel
- 8. Connector 13
- 9. Connector 12.



Terminal Blocks

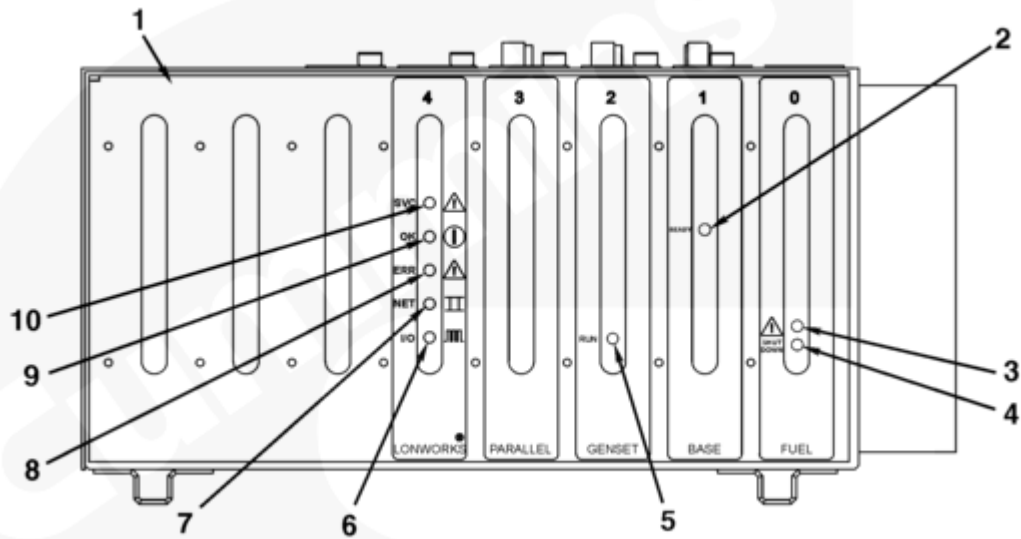
- 1. Connector 13 - modem or personal computer connector
- 2. TB3
- 3. Connector 12 - control panel update connection
- 4. Card cage
- 5. TB5
- 6. TB4
- 7. TB8
- 8. Optional common alarm relay - K10
- 9. Optional run relay - K11, K12, and K13.



19802784

ECM Card Cage

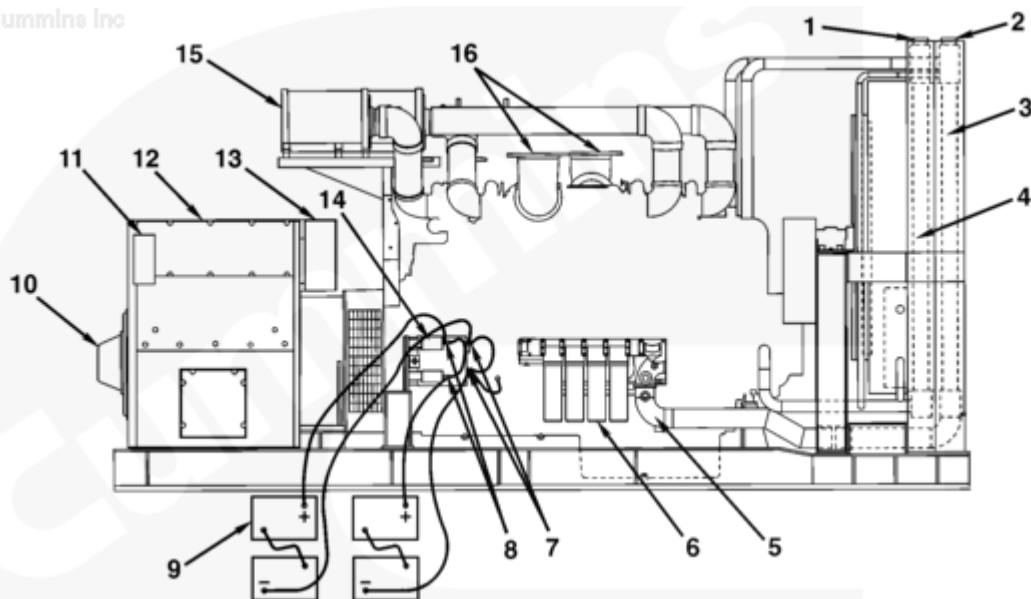
1. ECM card cage
2. Connector 04
3. Connector 01
4. Connector 06
5. Connector 05
6. Connector 02
7. Fuel board heat sink
8. Fuel board
9. Base board
10. Generator set board
11. Parallel board
12. LonWorks board
13. Delete plate
14. Backplane connector.



19802785

ECM Card Cage

1. ECM card cage
2. Ready light emitting diode
3. Warning light emitting diode
4. Shutdown light emitting diode
5. Run light emitting diode
6. Input/Output light emitting diode
7. Network light emitting diode
8. Error light emitting diode
9. OK light emitting diode
10. Service light emitting diode.

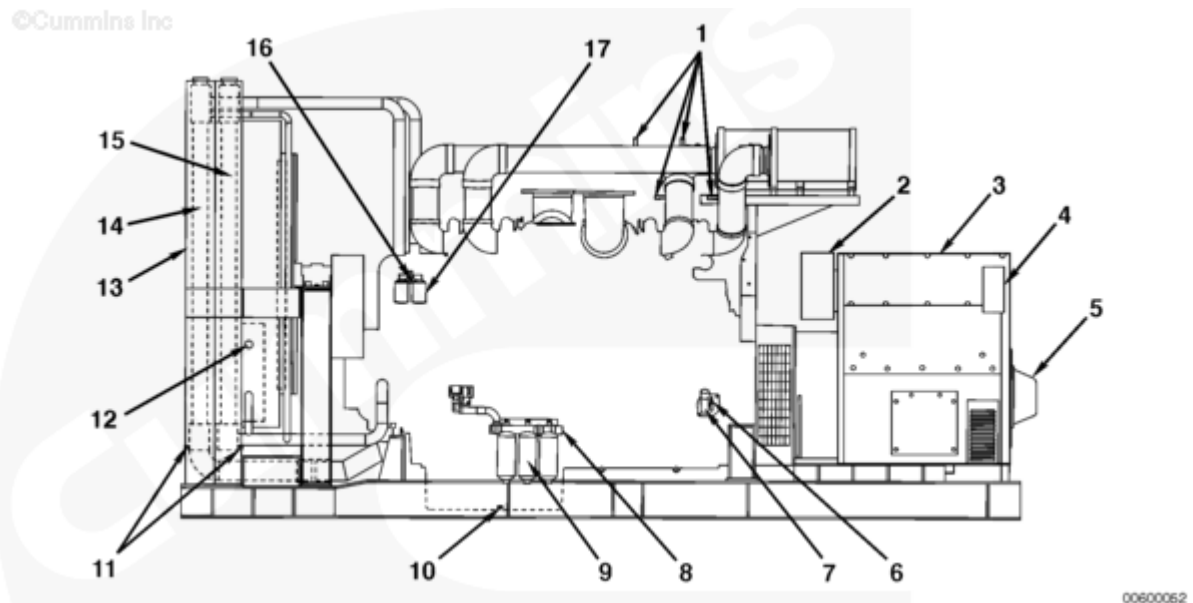


00600053

Generator Set (Right Side)

1. Radiator cap (LTA coolant fill location)

2. Radiator cap (jacket water coolant fill location)
3. Jacket water radiator
4. LTA radiator
5. Water inlet
6. Oil filters
7. Battery negative (-) connection
8. Battery positive (+) connection
9. Batteries
10. Alternator
11. Operator box
12. Alternator and control housing
13. Control box
14. Starters
15. Air cleaners
16. Exhaust outlet connections.



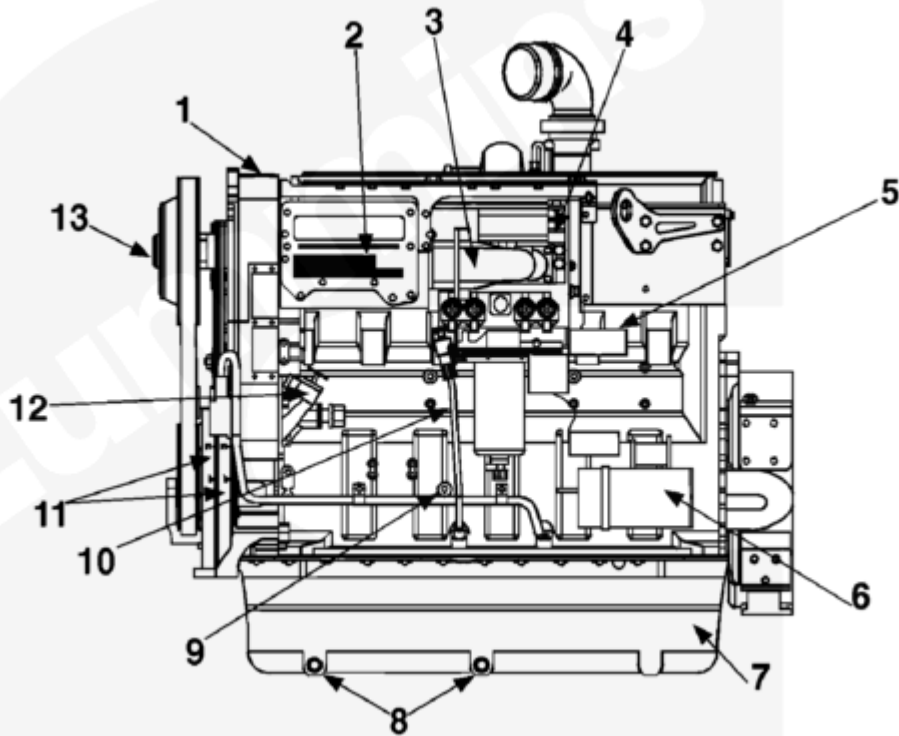
Generator Set (Left Side)

1. Restriction indications
2. Control box
3. Alternator control housing
4. Operator box
5. Alternator
6. Dipstick
7. Lubricating oil filter
8. Fuel inlet
9. Fuel filters
10. Oil drain
11. Coolant drain
12. Fuel spill from fuel cooler
13. Radiator

14. Jacket water radiator
15. LTA radiator
16. Coolant filter shutoff valve
17. Filters.

## Engine Views

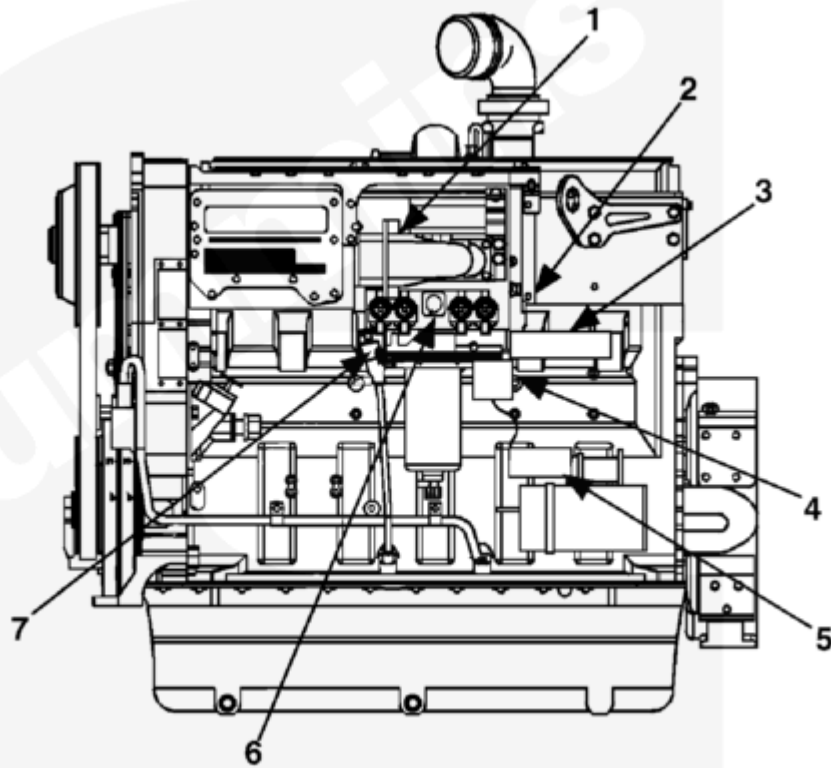
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19a00777

QSX15 - Intake Side (Power Generation)

1. Gear housing
2. Engine dataplate
3. Air intake
4. Fuel pump
5. Fuel filter
6. Starter
7. Lubricating oil pan
8. Lubricating oil drains
9. Crankshaft timing pin port
10. Lubricating oil dipstick
11. Vibration dampers
12. Lubrication oil fill/barring device
13. Fan hub.

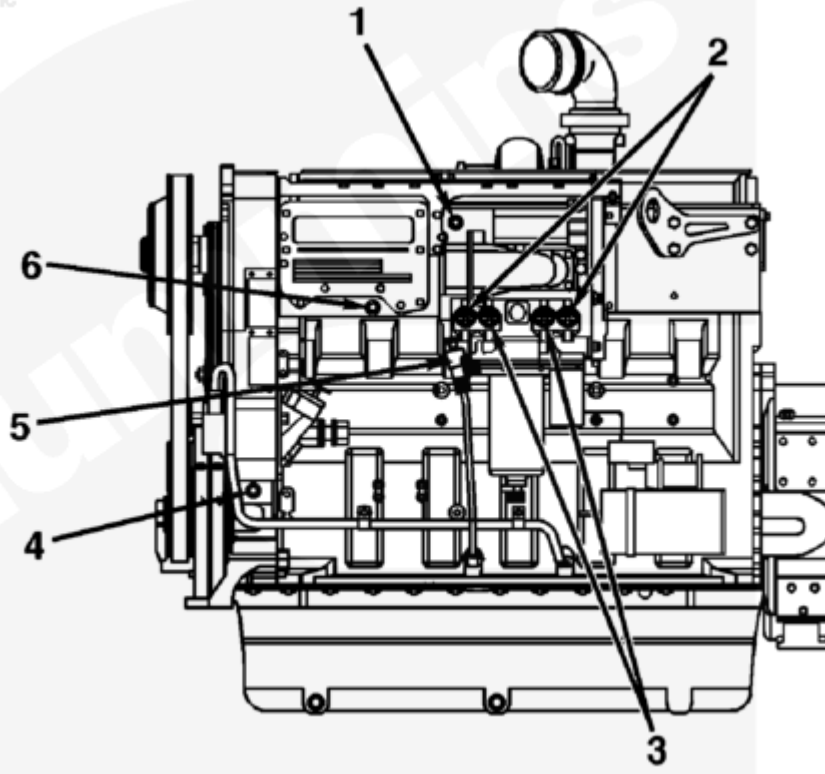


19a00774

QSX15 - Intake Side (Fuel System) Power Generation

1. Intake manifold pressure/temperature sensor
2. Fuel drain
3. Fuel inlet fitting/inlet filter housing
4. Fuel priming pump
5. Priming pump jumper harness/relay
6. Fuel shutoff valve
7. Oil pressure/temperature sensor.

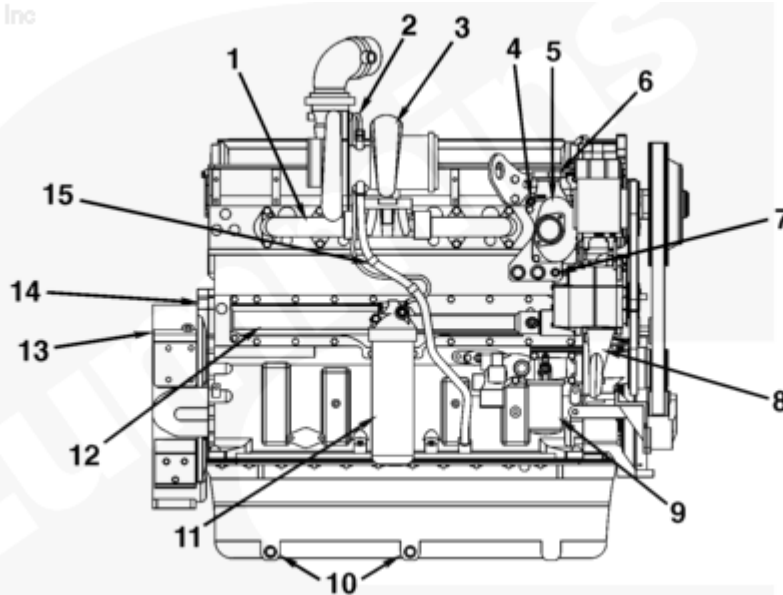




19a00775

QSX15 (Intake Side)

1. Camshaft position sensor
2. Fueling actuators
3. Timing actuators
4. Crankshaft position sensor
5. Fuel supply pressure sensor
6. Barometric pressure sensor.



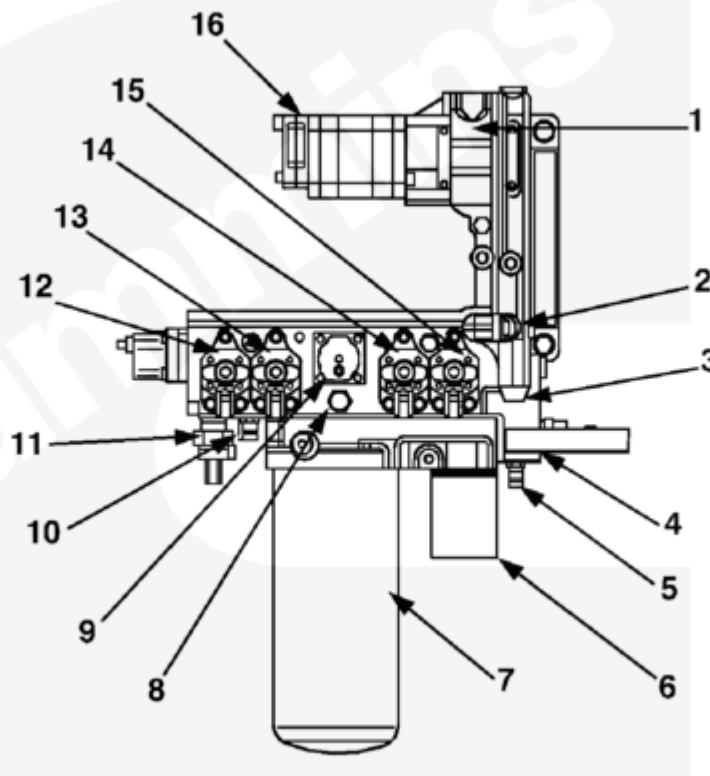
00800010

QSX15 (Exhaust Side)

1. Exhaust manifold

2. Lubricating oil inlet to turbocharger
3. Turbocharger
4. Thermostat housing vent
5. Thermostat housing
6. Engine coolant outlet
7. Coolant temperature sensor
8. Water pump
9. Coolant filter
10. Engine oil drain
11. Combination full-flow/bypass lubricating oil filter
12. Oil cooler
13. Flywheel housing
14. Engine serial number (ESN)
15. Turbocharger lubricating oil drain.

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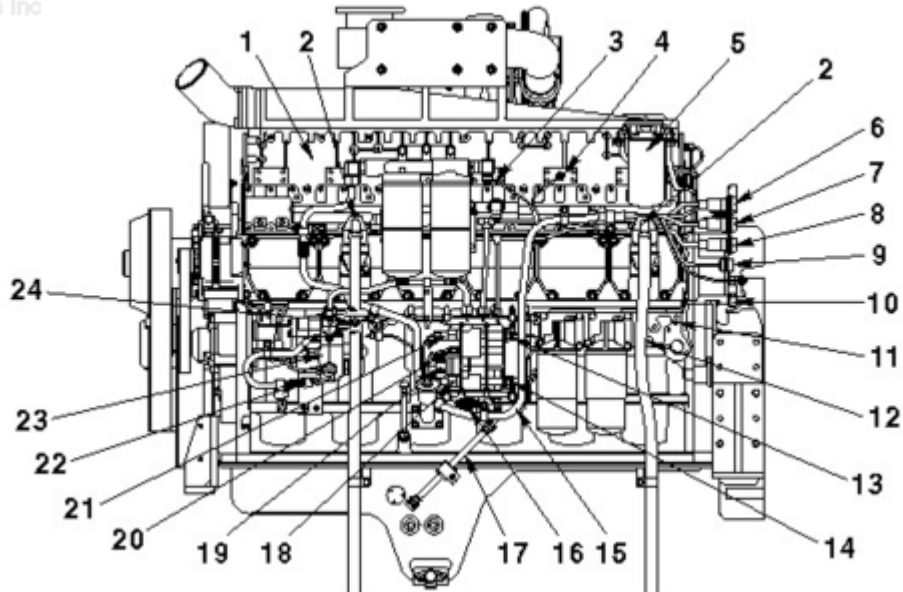
19a00776

QSX15 - Fuel Delivery Housing (Front View)

1. Priming bypass valve (internal)
2. 320-psi pressure regulator
3. 36-mc filter screen
4. Fuel inlet fitting/inlet filter housing
5. Quick disconnect pressure tap - suction side
6. Fuel priming pump
7. Fuel filter
8. 250-psi pressure regulator
9. Fuel shutoff valve
10. Quick-disconnect pressure tap - pressure side

11. Fuel pressure sensor
12. Front rail actuator
13. Front timing actuator
14. Rear timing actuator
15. Rear rail actuator
16. Fuel pump assembly.

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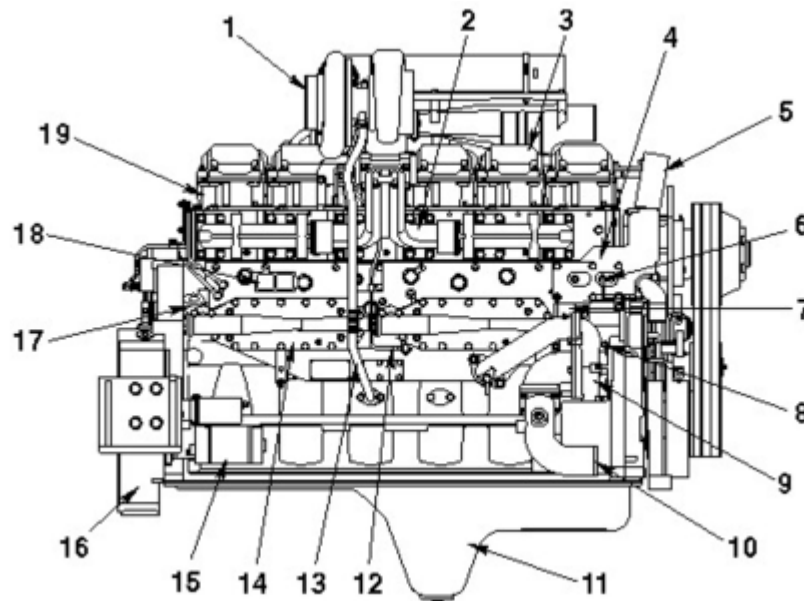
00400162

QSK23 (Left Side)

1. Intake manifold
2. Crankcase breathers
3. Intake manifold pressure sensor
4. Intake manifold temperature sensor
5. Coolant filter
6. Inline B connector
7. Inline A connector
8. Inline C connector
9. Datalink connector
10. Engine speed sensor
11. Oil pressure check point (after filters)
12. Oil pressure check point (before filters)
13. Fuel timing pressure sensor
14. Fuel rail pressure sensor
15. Wiring harness
16. Barometric pressure sensor
17. Fuel rail dampening hose
18. Fuel rail actuator
19. Fuel shutoff valve
20. Fuel temperature sensor
21. Fuel timing actuator
22. Fuel pump pressure sensor

- 23. Fuel pump actuator
- 24. Oil pressure sensor.

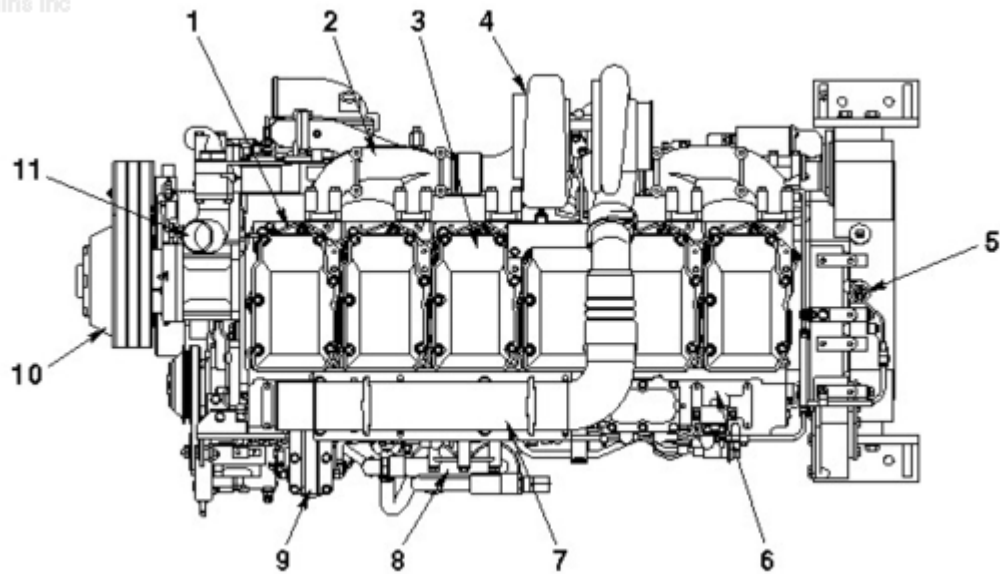
©Cummins Inc



00400165

QSK 23 (Right Side)

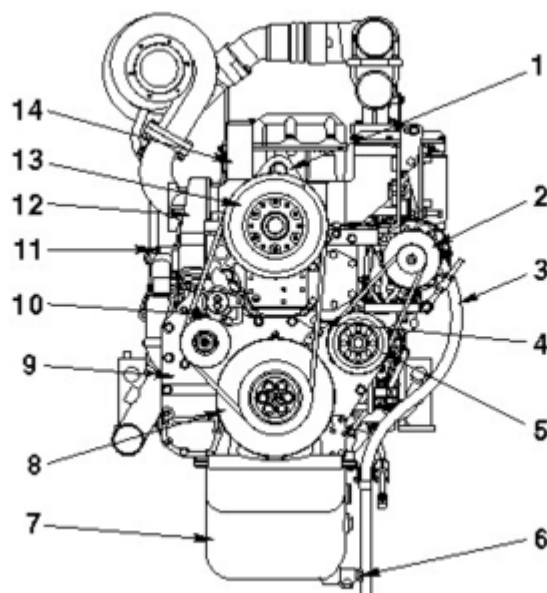
- 1. Turbocharger
- 2. Exhaust manifold (3-piece)
- 3. Rocker lever cover
- 4. Thermostat housing
- 5. Engine coolant outlet
- 6. Coolant temperature sensor
- 7. Coolant bypass tube
- 8. Oil pressure checkpoint (piston cooling nozzle gallery)
- 9. Water pump
- 10. Engine coolant inlet
- 11. Oil pan
- 12. Turbocharger oil supply line
- 13. Turbocharger oil drain line
- 14. Oil cooler housing
- 15. Starter
- 16. Flywheel housing
- 17. Oil pressure checkpoint (after oil coolers)
- 18. Engine dataplate
- 19. Rocker lever housing.



00400166

QSK23 (Top View)

1. Coolant vent tube
2. Exhaust manifold
3. Rocker lever cover
4. Turbocharger
5. Engine speed sensor (industrial **only**)
6. Intake manifold
7. Charge air piping (turbocharger to charger air cooler)
8. Fuel filter head
9. Coolant filter head (industrial **only**)
10. Fan hub assembly
11. Engine coolant outlet.

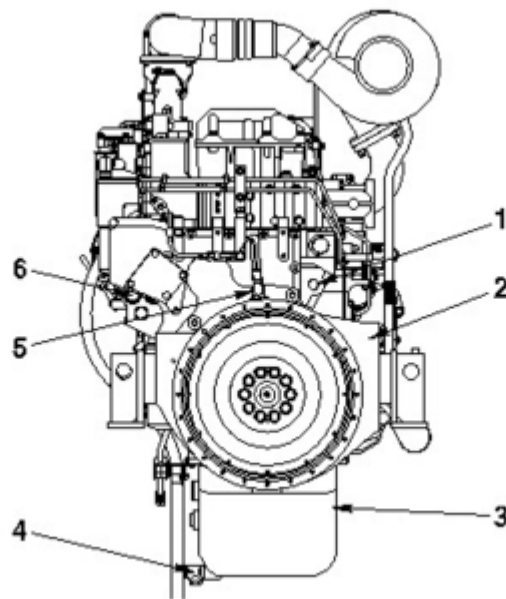


00400167

QSK23 (Front View)

1. Front engine lifting bracket
2. Alternator
3. Blowby tube
4. Cooling fan drive belt
5. Drive pulley (for alternator and refrigerant compressor)
6. Oil drain valve
7. Oil pan
8. Vibration damper
9. Front gear cover
10. Cooling fan belt tensioner (industrial **only**)
11. Coolant temperature sensor
12. Thermostat housing
13. Fan hub assembly
14. Engine coolant outlet.

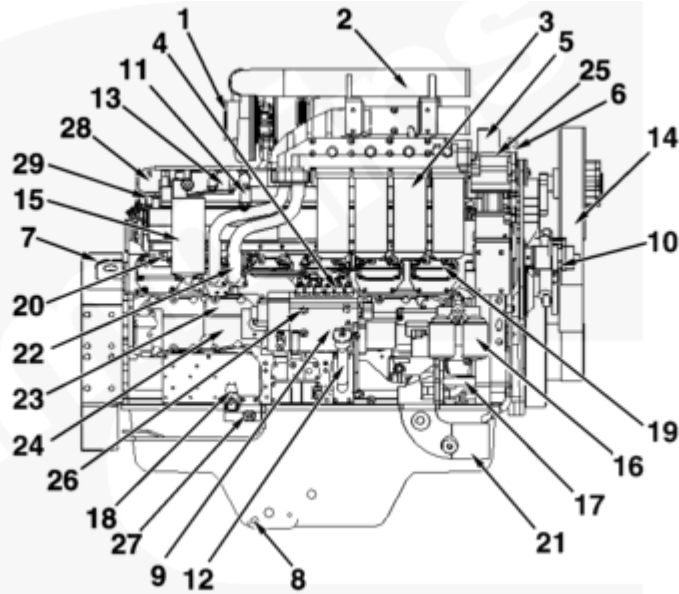
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00400168

QSK23 (Rear View)

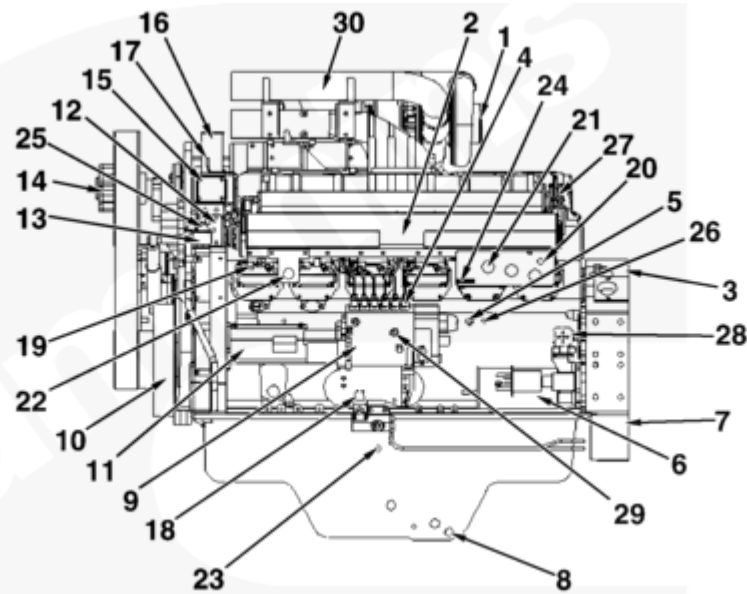
1. Oil pressure check point (after oil coolers)
2. Flywheel housing
3. Oil pan
4. Engine oil drain valve
5. Engine speed sensor (industrial **only**)
6. OEM electrical connections (industrial **only**)



19803650

QST30 GCS (Right Side)

1. Turbocharger inlet connection
2. ATA crossover connection
3. Combo lube oil filters
4. High-pressure fuel lines
5. Coolant outlet connection
6. Lifting bracket
7. Flywheel housing
8. Oil drain port
9. Right bank fuel pump
10. Fan belt idler assembly
11. Fuel inlet fitting
12. Oil filler tube
13. Fuel supply pump housing
14. Fan hub
15. Fuel filter
16. Coolant additive
17. Coolant pump
18. Fuel shutoff valve 2 (right bank)
19. Intake manifold pressure sensor 2 (right bank)
20. Intake manifold temperature sensor 2 (right bank)
21. Coolant inlet connection
22. Oil transfer tubes
23. Centinel driver connector
24. Centinel oil tank level switch connector
25. Alternator
26. Overflow valve
27. Fuel outlet fitting
28. Fuel supply pressure sensor
29. Fuel supply temperature sensor



19803651

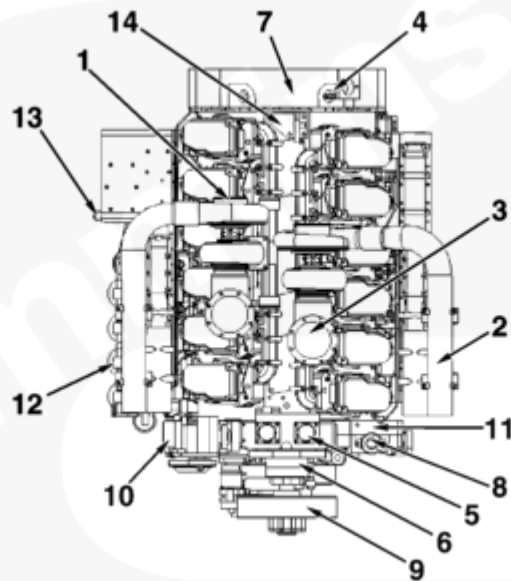
QST30 GCS - (Left Side)

1. Turbocharger inlet connection
2. Aftercooler housing/intake manifold
3. Engine speed sensor
4. High pressure fuel supply lines
5. Oil pressure sensor
6. Starter
7. Flywheel housing
8. Oil drain port
9. Left bank fuel pump
10. Vibration damper
11. Fuel pump drive housing
12. Coolant temperature sensor
13. Crankcase breather
14. Fan hub
15. Thermostat housing
16. Coolant outlet connection
17. Coolant vent tube
18. Fuel shutoff valve 1 (left bank)
19. Intake manifold pressure sensor 1 (left bank)
20. Intake manifold temperature sensor 1 (left bank)
21. INLINE connectors
22. Datalink connector
23. Oil level sensor (optional)
24. Barometric pressure sensor
25. Coolant pressure sensor
26. Oil temperature sensor
27. Heater connector
28. Starter solenoid
29. Overflow valve
30. Turbocharger housing



### 30. ATA crossover connection.

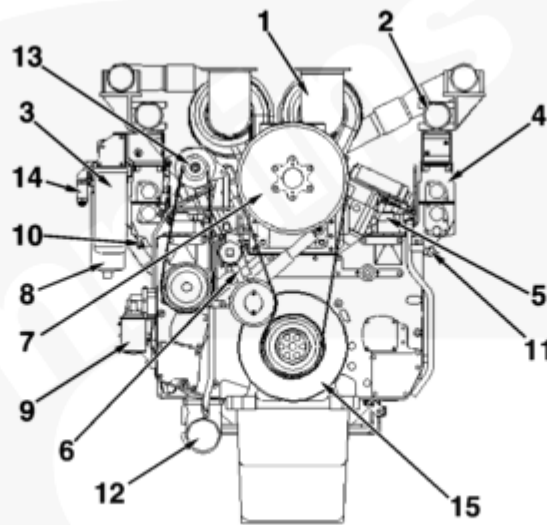
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19803652

QST30 GCS - (Top View)

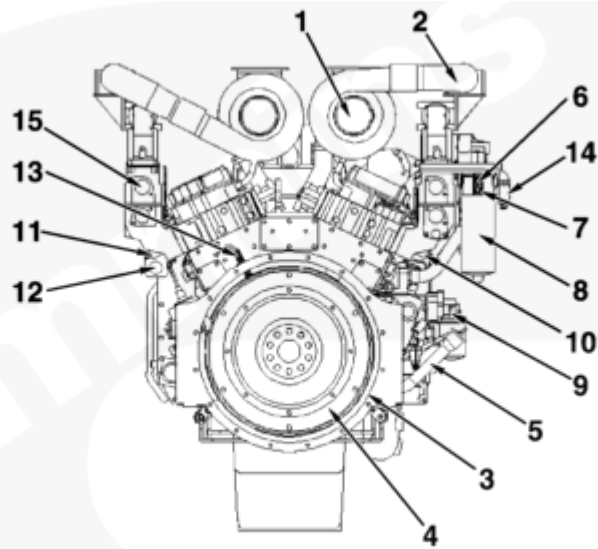
1. Turbocharger inlet connection
2. ATA crossover connection
3. Exhaust outlet connection
4. Engine speed sensor
5. Coolant outlet connection
6. Front lifting bracket
7. Flywheel housing
8. Crankcase breather
9. Fan hub
10. Alternator
11. Coolant level switch connector
12. Combo lube oil filters
13. Fuel inlet fitting
14. Rear lifting bracket.



19803653

QST30 GCS (Front View)

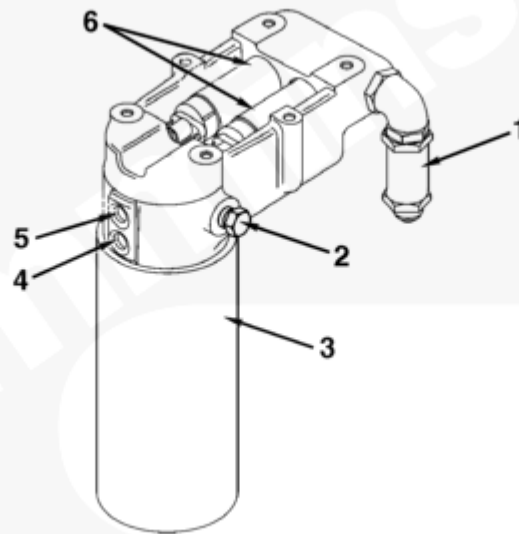
1. Turbocharger exhaust connection
2. ATA crossover connection
3. Combo lube oil filters
4. Aftercooler housing/intake manifold
5. Crankcase breather
6. Fan belt Idler assembly
7. Fan hub
8. Fuel filter
9. Coolant additive
10. Intake manifold pressure sensor 2 (right bank)
11. Intake manifold pressure sensor 1 (left bank)
12. Coolant inlet connection
13. Alternator
14. Fuel inlet fitting
15. Vibration damper.



19803654

QST30 GCS (Rear View)

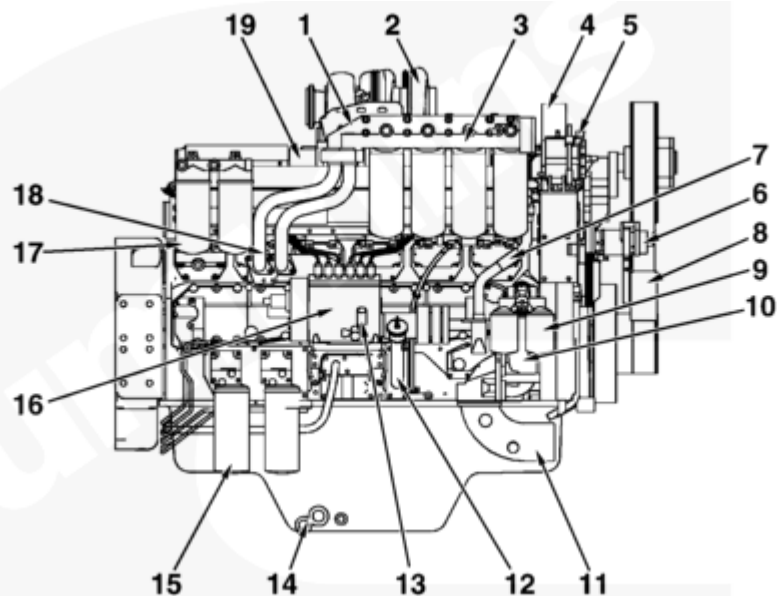
1. Turbocharger inlet connection
2. ATA crossover tube
3. Flywheel housing
4. Flywheel
5. Oil filler tube
6. Fuel supply pressure sensor
7. Fuel supply temperature sensor
8. Fuel filter
9. Coolant additive
10. Intake manifold temperature sensor 2 (right bank)
11. Intake manifold temperature sensor 1 (left bank)
12. Inline-E connector
13. Engine speed sensor
14. Fuel inlet fitting
15. Aftercooler housing/intake manifold.



19803660

QST30 (Fuel Supply Pump and Filter)

1. Fuel supply connection
2. Fuel outlet connection
3. Fuel filter
4. Fuel supply temperature sensor
5. Fuel supply pressure sensor
6. Fuel supply pump 1 and 2.



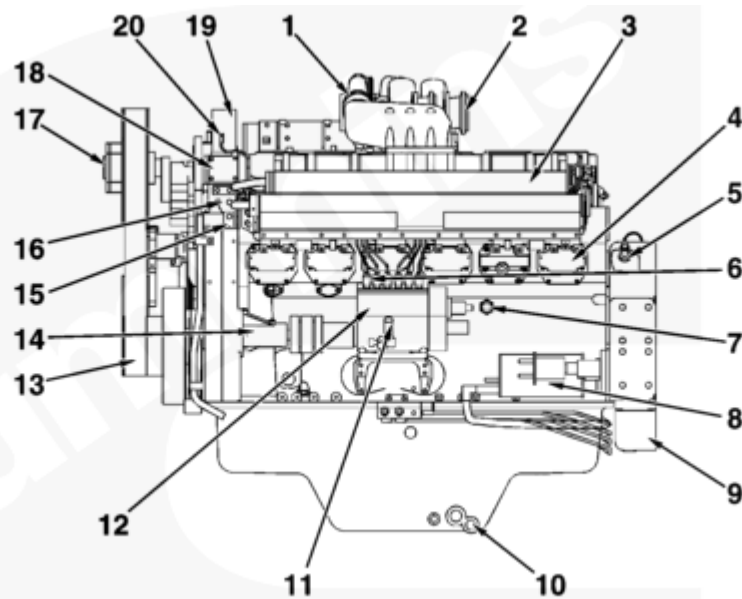
19803655

QST30 Upgrade (Right Bank)

1. Air crossover housing
2. Turbocharger
3. Full flow oil filters
4. Water outlet connection
5. Lifting bracket

6. Fan belt idler assembly
7. Water bypass tube
8. Fan belt Idler pulley
9. Water filter
10. Water pump
11. Water inlet connection
12. Oil filler tube
13. Lift pump
14. Oil drain
15. Fuel filter
16. Fuel pump
17. Oil bypass filters
18. Oil transfer tube
19. Intake manifold.

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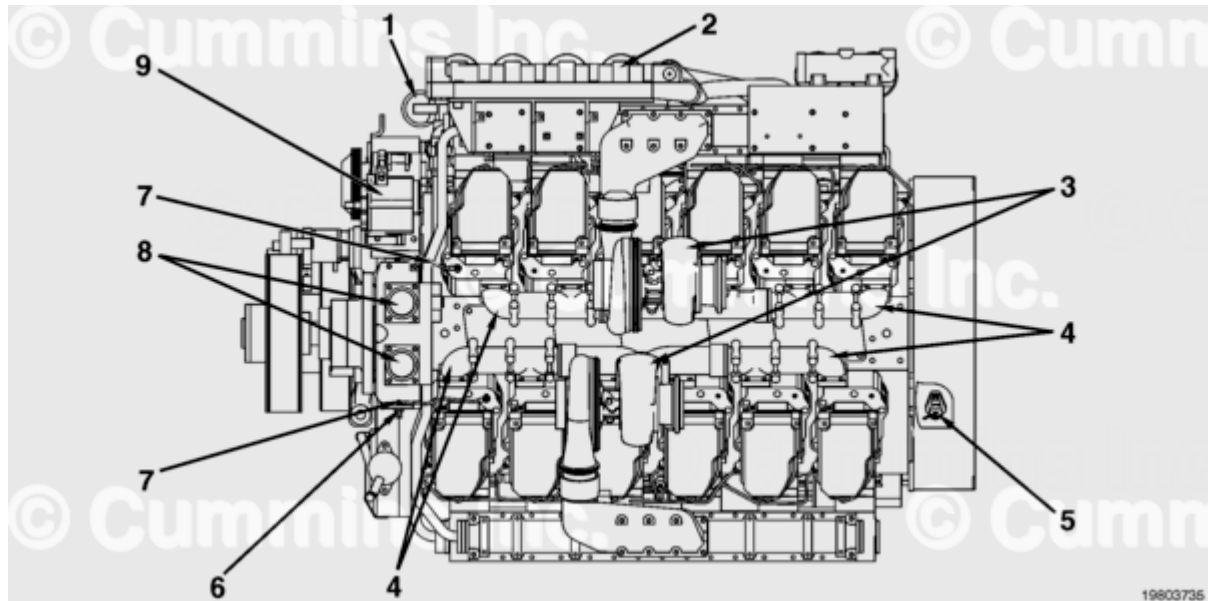


19803656

QST30 Upgrade (Left Bank)

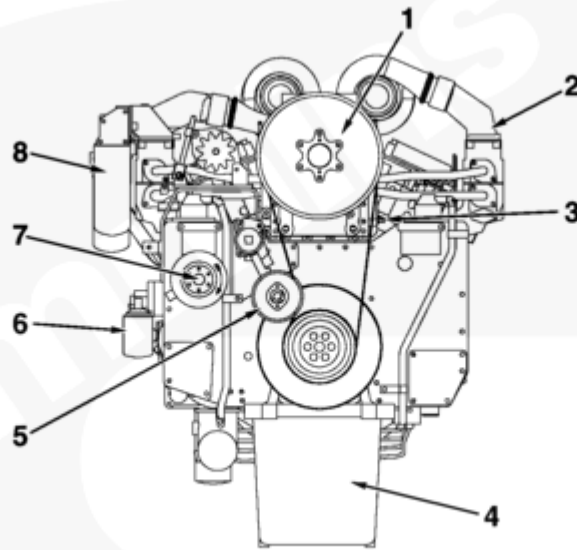
1. Turbocharger inlet connection
2. Turbocharger outlet connection
3. Aftercooler housing
4. Cam follower cover
5. Engine speed sensor
6. High pressure fuel supply lines
7. Oil pressure sensor
8. Starter
9. Flywheel housing
10. Oil drain
11. Fuel lift pump
12. Fuel pump
13. Vibration damper
14. Fuel pump drive

15. Coolant temperature sensor
16. Crankcase breather
17. Fan hub
18. Thermostat housing
19. Water outlet connection
20. Water vent tubes.



QST30 Upgrade (Top View)

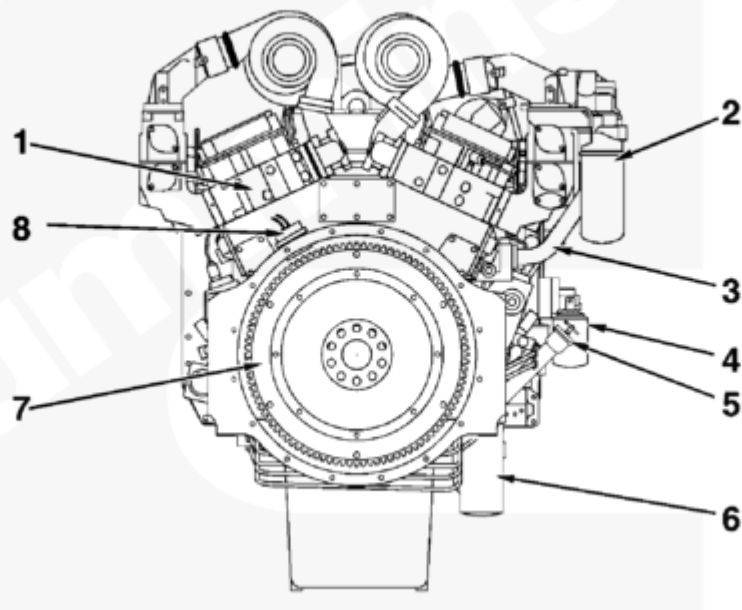
1. Water filter
2. Full flow oil filters
3. Turbochargers
4. Exhaust manifolds
5. Engine speed sensor
6. Coolant temperature sensor
7. Water vent connection
8. Water outlet connection
9. Alternator.



19803658

QST30 Upgrade (Front View)

1. Fan hub
2. Air crossover
3. Coolant temperature sensor
4. Oil pan
5. Fan idler tensioner
6. Water filters
7. Engine barring location
8. Full flow oil filters.



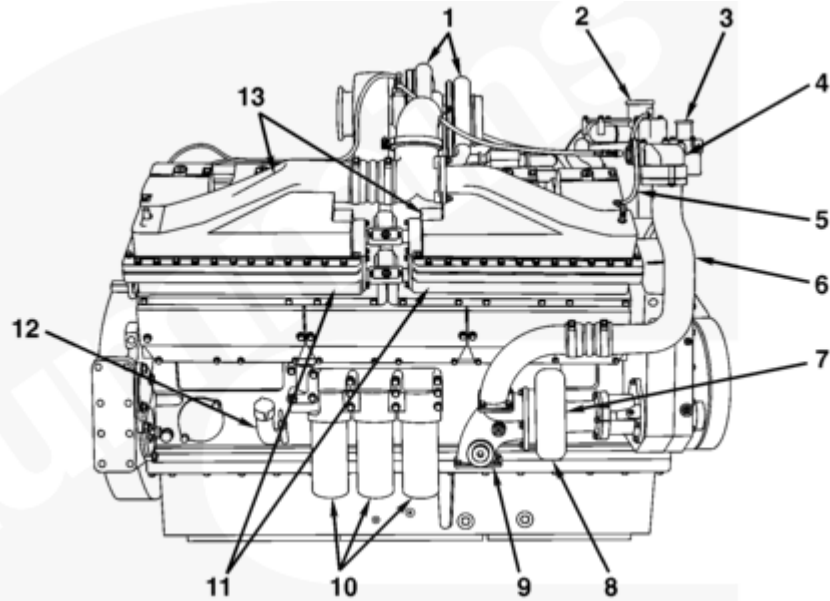
19803736

QST30 Upgrade (Rear View)

1. Cylinder head

2. Bypass filters
3. Oil transfer tube
4. Water filters
5. Oil fill
6. Fuel filters
7. Flywheel
8. Engine speed sensor.

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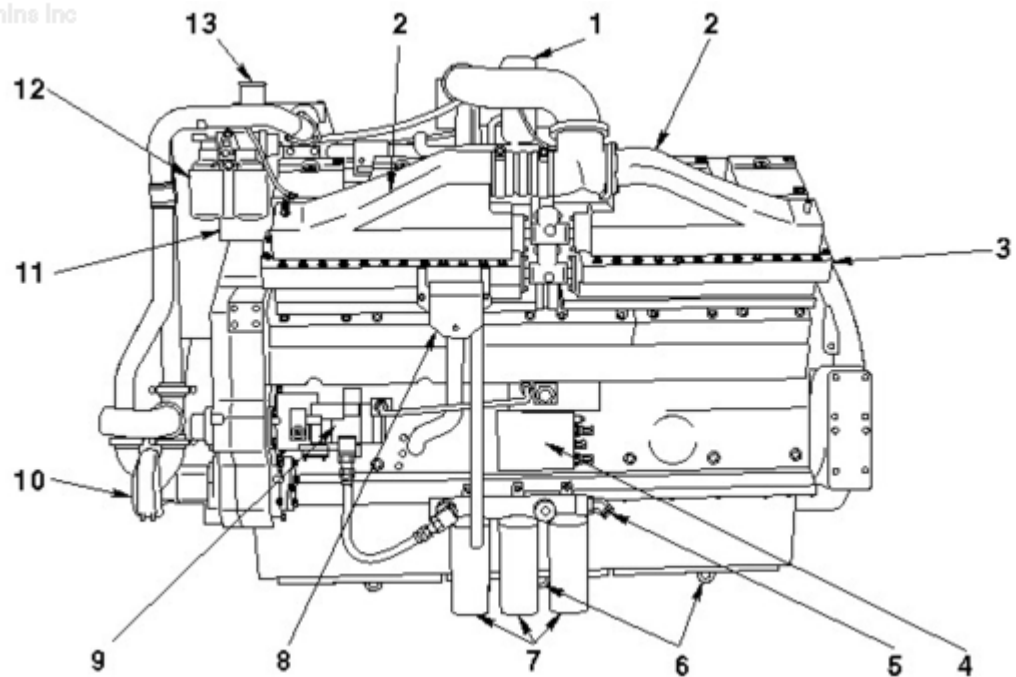


00600047

QSK45 (Right Side)

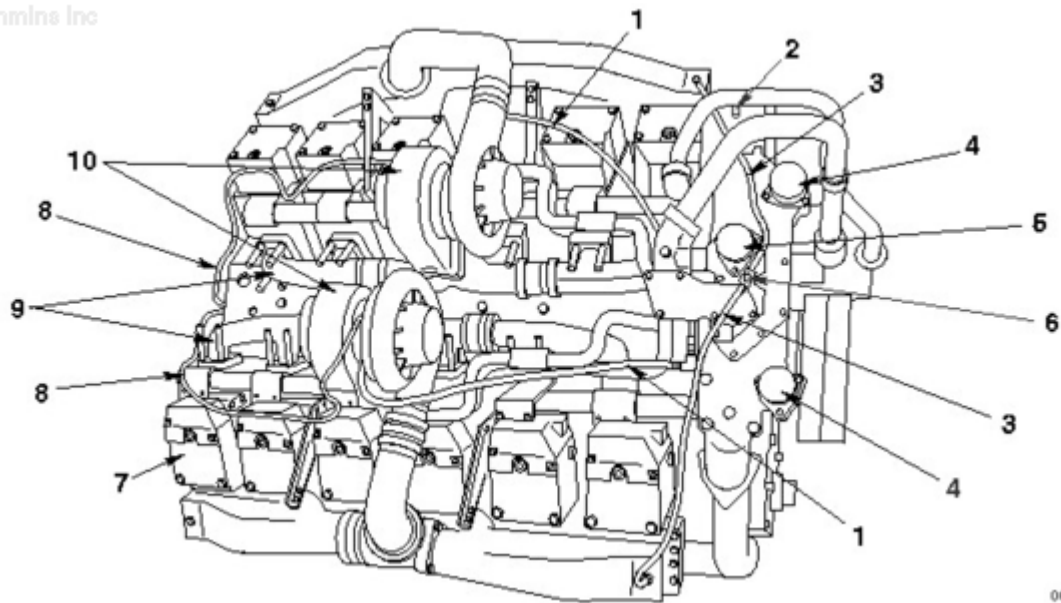
1. Turbochargers
2. LTA water outlet connection
3. Engine water outlet connections
4. Thermostat housing
5. Aftercooler vent line
6. Water bypass tube
7. Water pump
8. Water pump drain
9. Coolant inlet connection
10. Oil filters
11. Aftercooler assemblies
12. Oil fill and dipstick tubes (location can vary)
13. Intake manifold cover.





QSK45 (Left Side)

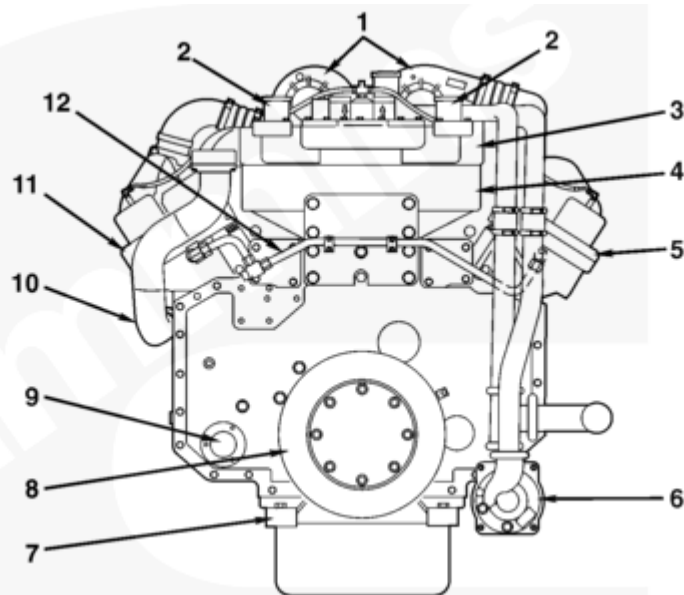
1. Turbocharger
2. Intake manifold covers
3. Aftercooler assemblies
4. Electronic control valve assembly (ECVA)
5. Fuel inlet connection
6. Engine oil drains
7. Fuel filters
8. Crankcase breather
9. Fuel pump
10. LTA water pump
11. Thermostat housing support
12. Water filters
13. LTA water outlet connection.



00600045

QSK45 (Top View)

1. Coolant supply lines to turbocharger
2. Aftercooler water inlet temperature sensor
3. Coolant vent connection aftercoolers
4. Engine water outlet connection
5. LTA water outlet connection
6. Aftercooler vent line connection engine
7. Rocker lever cover
8. Oil supply lines to turbocharger
9. Exhaust manifolds
10. Turbochargers.



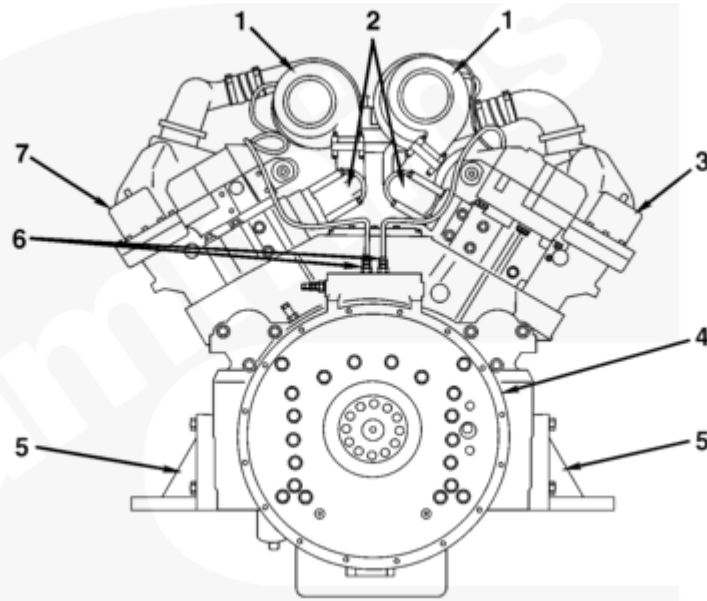
00600048

QSK45 (Front View)

1. Turbochargers

2. Engine water outlet connections
3. Jacket water radiator
4. LTA radiator
5. Left-bank aftercooler assembly
6. LTA water pump
7. Front engine mount
8. Vibration damper
9. Alternator drive
10. Water bypass tube
11. Right-bank aftercooler assembly
12. Fuel return line.

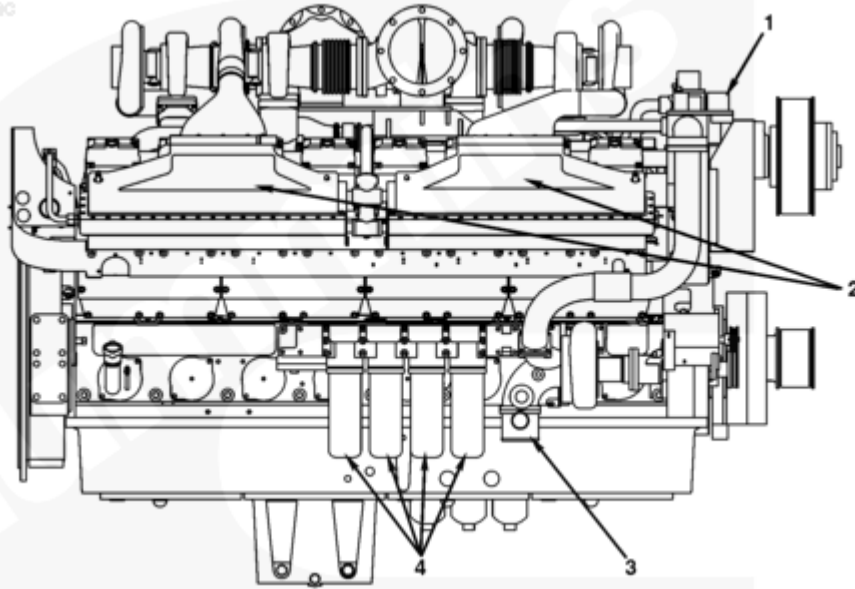
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00600049

QSK45 (Rear View)

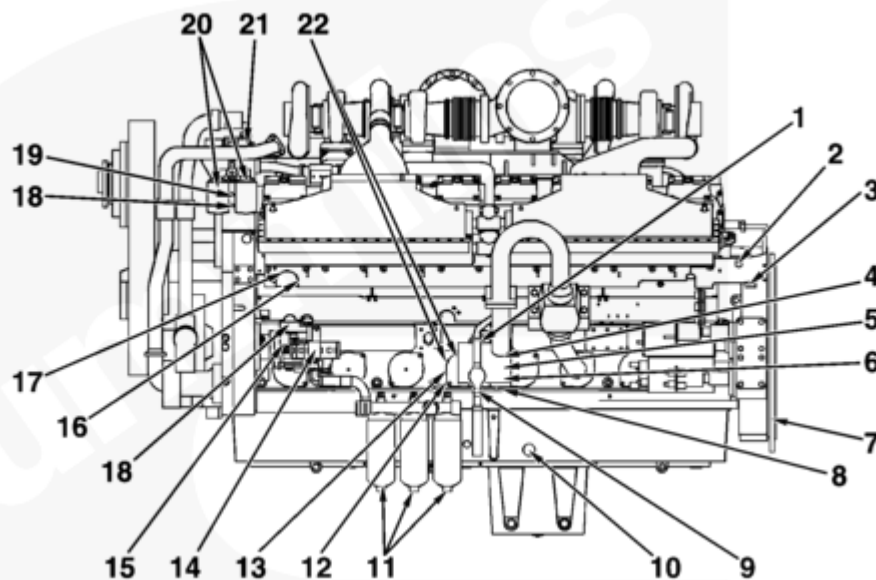
1. Turbochargers
2. Exhaust manifolds
3. Right-bank aftercooler assembly
4. Flywheel housing
5. Rear engine mount
6. Turbocharger oil supply lines
7. Left-bank aftercooler assembly.



19802616

QSK60 (QSK45 similar) - Right Bank

1. Coolant outlet
2. Aftercooler assembly
3. Coolant inlet
4. Oil filters.



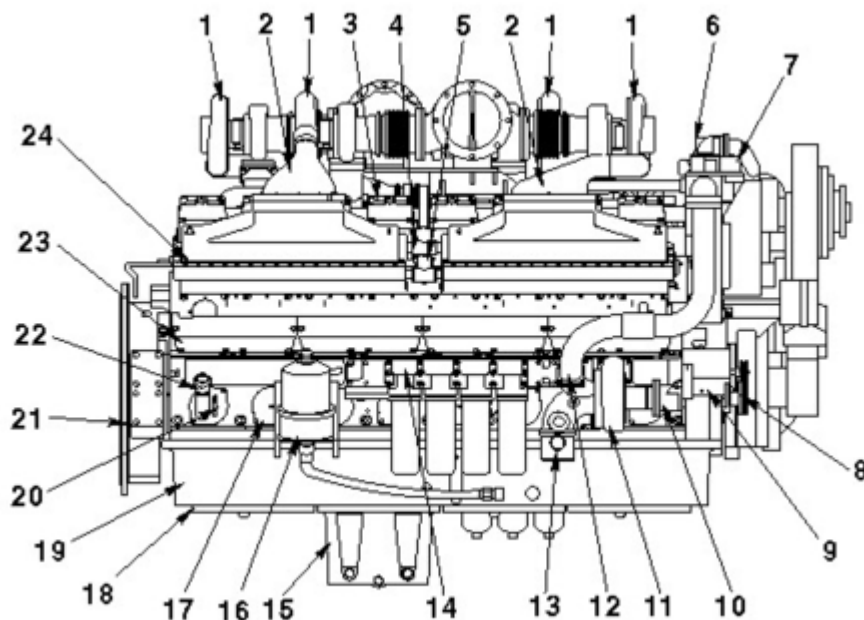
19803661

QSK60 (QSK45 similar) - Left Bank

1. Control valve body
2. Oil pressure sensor
3. Engine speed sensor
4. Timing rail pressure sensor
5. Barometric pressure sensor
6. Fuel temperature sensor
7. Flywheel housing

8. Fuel rail pressure sensor
9. Blowby pressure sensor
10. Oil level sensor
11. Fuel filters
12. Fuel rail actuator
13. Fuel shutoff valve
14. Fuel pump
15. Fuel pump output pressure Compuchek® fitting
16. Intake manifold pressure sensor
17. Intake manifold temperature sensor
18. Coolant pressure sensor
19. Coolant temperature sensor
20. Coolant filters
21. Aftercooler water temperature sensor
22. Timing rail actuators.

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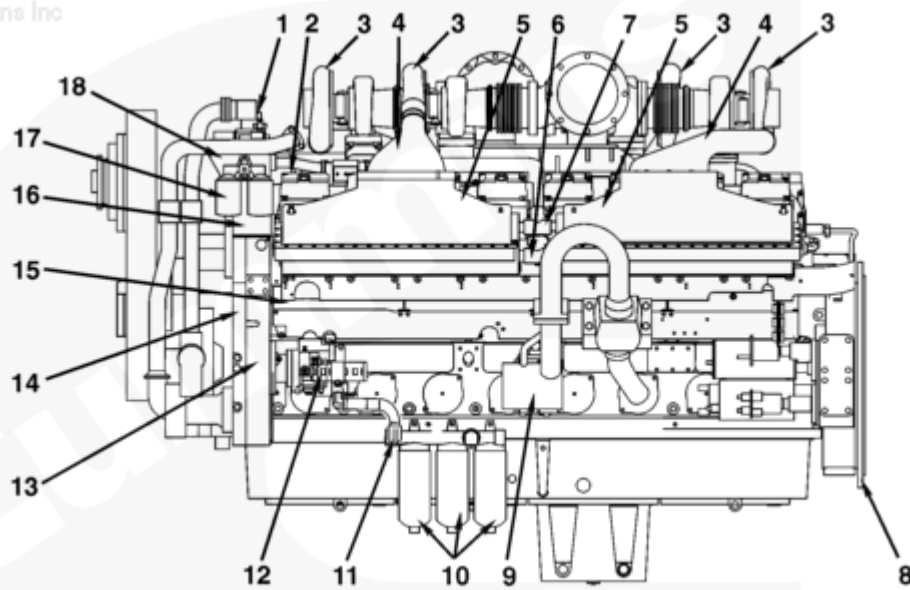
00600018

QSK60 (Right Side)

1. Turbochargers
2. Air crossover connections
3. Rocker lever cover
4. Aftercooler water outlet tube
5. Aftercooler water inlet tube
6. LTA water outlet connection
7. Engine water outlet connections
8. Alternator drive
9. Alternator
10. Water pump drive
11. Water pump

12. Water bypass tube
13. Water inlet connection
14. Oil filter head
15. Oil pan sump
16. Centrifuge bypass filter (optional)
17. Handhole covers
18. Oil pan adapter plate
19. Oil pan adapter
20. Dipstick (location can vary)
21. Flywheel housing
22. Oil fill tube (location can vary)
23. Cam follower cover
24. Aftercooler assembly.

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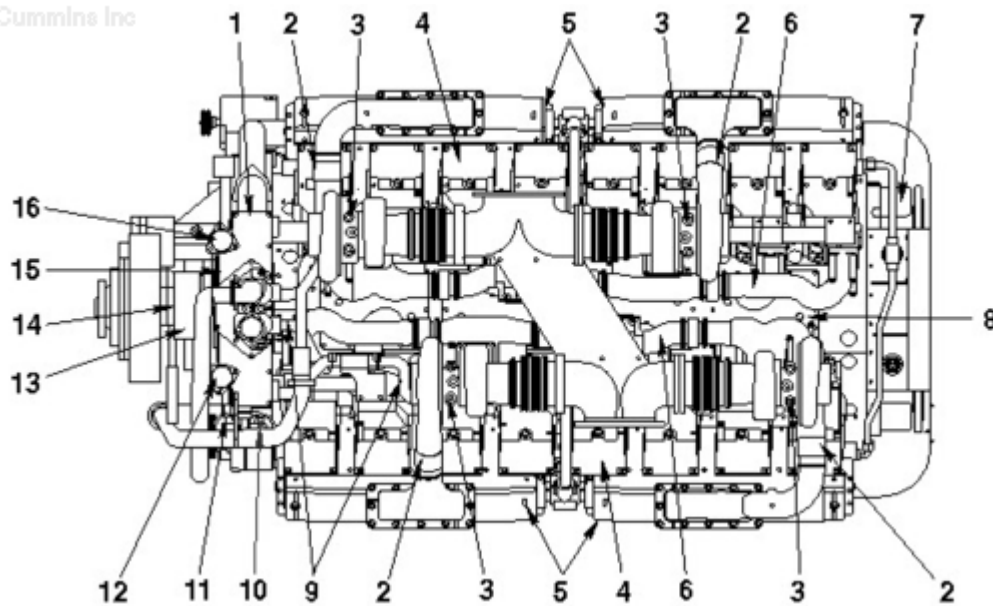
00600050

QSK60 (Left Side)

1. LTA water outlet connections
2. Rocker lever cover
3. Turbochargers
4. Air crossover connection
5. Aftercooler assemblies
6. Aftercooler water inlet tube
7. Aftercooler water outlet tube
8. Flywheel housing
9. Electronic control valve assembly
10. Fuel filter
11. Fuel filter head
12. Fuel pump
13. Front gear housing
14. Front gear cover
15. Cam follower cover

- 16. Thermostat housing support
- 17. Coolant filters
- 18. Thermostat housing.

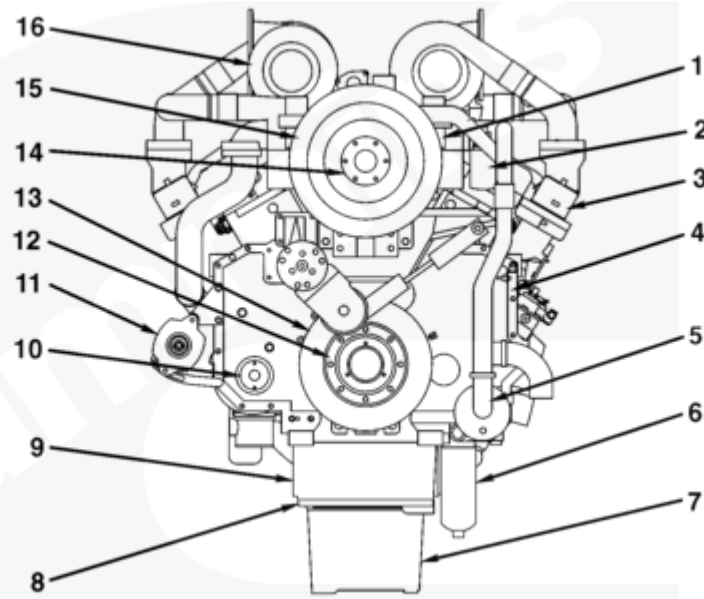
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00600046

QSK60 (Top View)

- 1. Thermostat housing
- 2. Air crossover connections
- 3. Turbochargers
- 4. Rocker lever cover
- 5. Aftercooler assemblies
- 6. Exhaust manifolds
- 7. Flywheel housing
- 8. Oil cooler cover plate
- 9. Aftercooler water tube
- 10. Aftercooler water inlet temperature sensor
- 11. Coolant filter head
- 12. Left-bank water outlet
- 13. Fan hub support
- 14. Fan hub shaft
- 15. Thermostat housing Support
- 16. Right-bank water outlet.

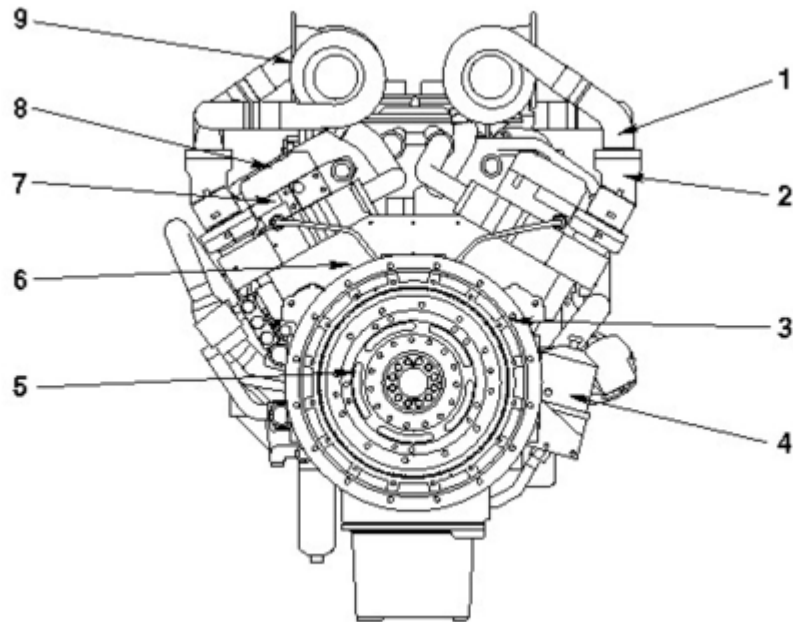


00600051

QSK60 (Front View)

1. Thermostat housing
2. Coolant filter
3. Aftercooler assembly
4. Gear cover
5. LTA water pump
6. Fuel filter
7. Oil pan sump
8. Oil pan adapter plate
9. Oil pan adapter
10. Charging alternator drive pulley
11. Charging alternator
12. Crankshaft pulley
13. Vibration dampers
14. Fan hub
15. Fan hub pulley
16. Turbocharger.

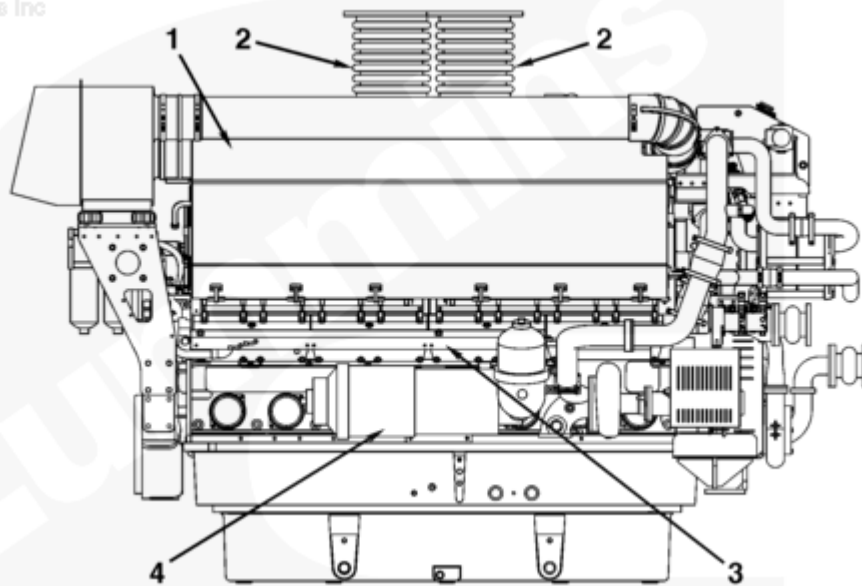




00600022

QSK60 (Rear View)

1. Air crossover connector
2. Aftercooler assembly
3. Flywheel housing
4. Centrifuge bypass filter
5. Flywheel
6. Cylinder block
7. Rocker lever housing
8. Rocker lever cover
9. Turbocharger.

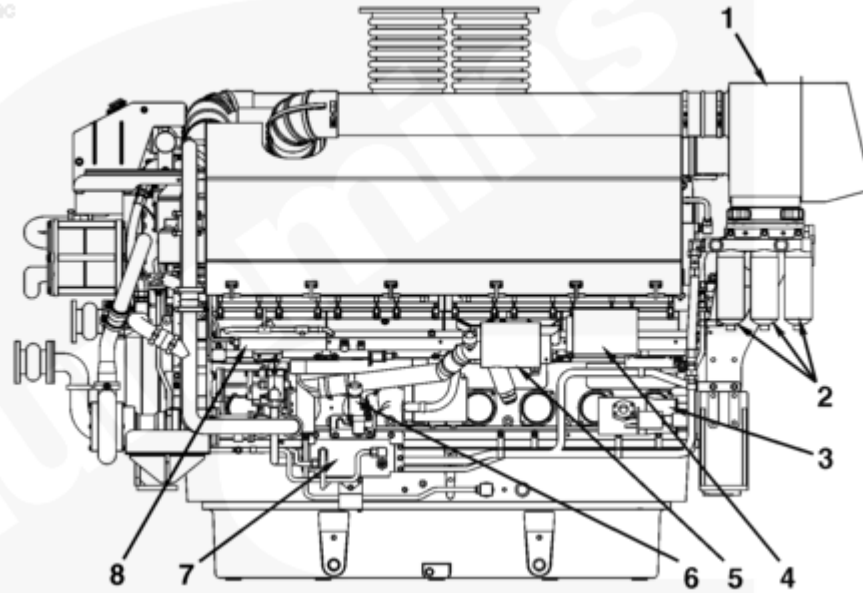


00600215

QSK60 Marine - Right Side View

1. Aftercooler heat shield
2. Exhaust outlet bellows
3. Right-bank wiring harness
4. Eliminator lubricating oil filter assembly.

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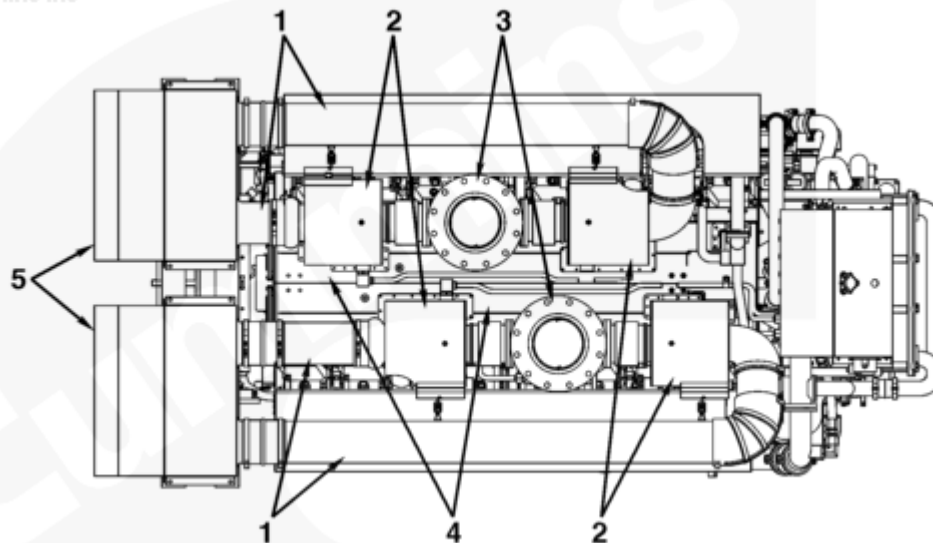


00600216

QSK60 Marine - Left Side View

1. Air cleaner assembly
2. Fuel filters
3. Starter
4. Electronic control module (ECM)
5. Crankcase breather
6. Oil fill and dipstick tubes
7. Fuel cooler (optional)
8. Left-bank wiring harness

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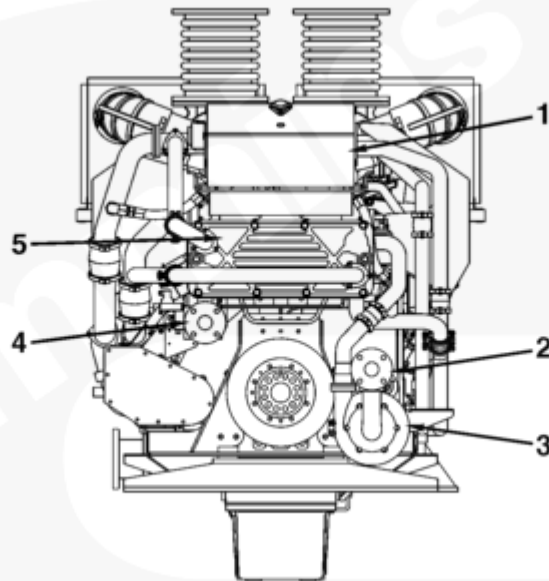


00600217

QSK60 Marine - Top View

1. Air intake tubes
2. Turbocharger heat shields
3. Exhaust outlet connections
4. Exhaust manifold heat shields
5. Air cleaner assemblies.

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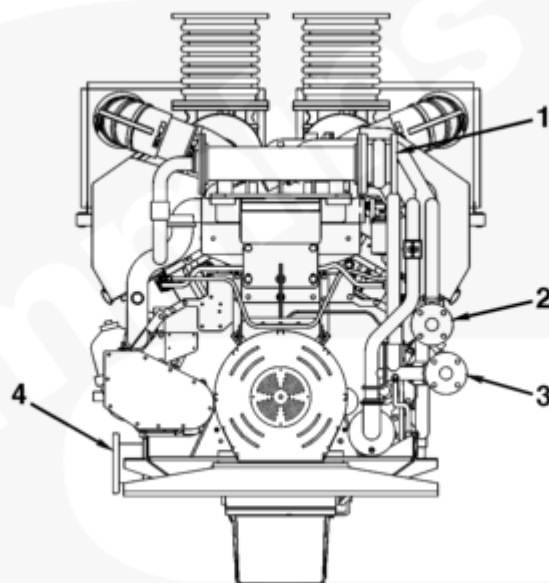


00600213

QSK60 Marine Heat Exchanger Cooled - Front View

1. Expansion tank
2. Seawater inlet connection
3. Seawater pump
4. Seawater outlet connection
5. Heat exchanger.

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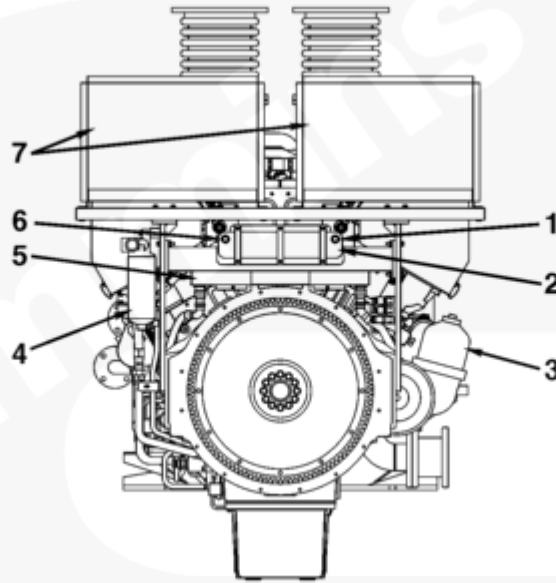


00600212

QSK60 Marine Keel Cooled - Front View

1. Engine coolant outlet connection to main engine cooler
2. LTA coolant outlet connection to LTA cooler
3. LTA coolant inlet connection from LTA cooler
4. Engine coolant inlet connection from main engine cooler.

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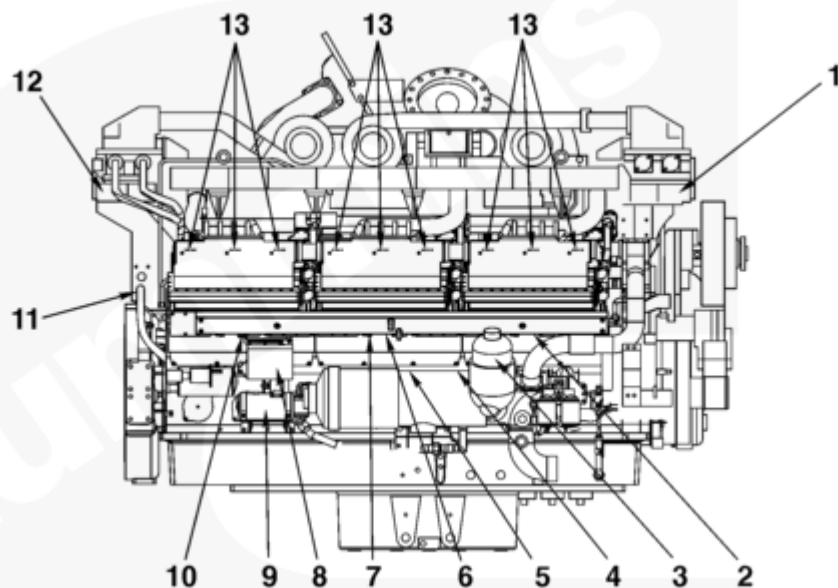


00600214

QSK60 Marine - Rear View

1. Gear oil inlet or outlet connection
2. Gear oil cooler (optional)
3. Eliminator lubricating oil filter assembly
4. Fuel filter
5. Rear engine harness
6. Gear oil inlet or outlet connection
7. Air filter assemblies.

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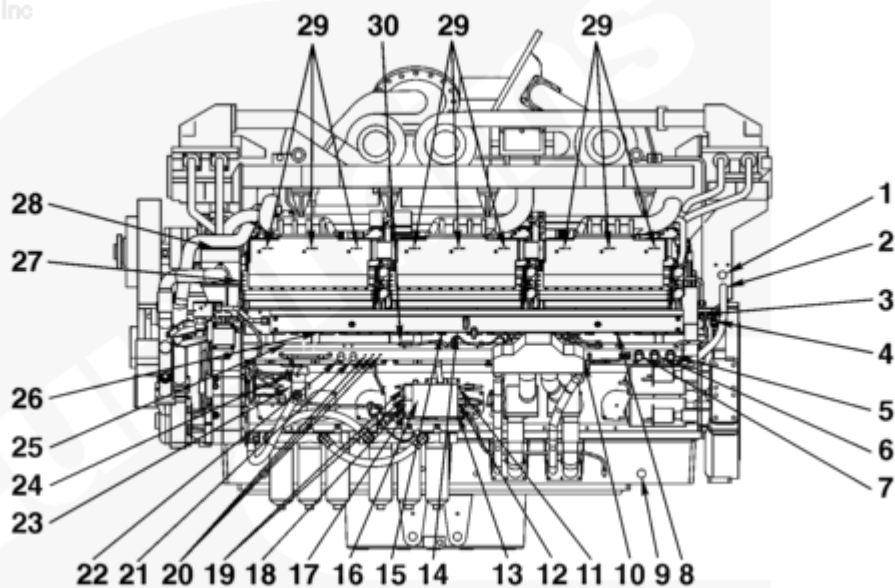


00600207

QSK78 (Right Side)

1. Front interstage pressure sensor
2. Right bank front intake manifold temperature sensor
3. Eliminator centrifuge speed sensor
4. Pre-oil filter pressure sensor
5. Post oil filter pressure sensor
6. Right bank middle intake manifold temperature sensor
7. Right bank intake manifold pressure sensor
8. CENSE™ ECM
9. Prelubricating oil pump
10. Right bank rear intake manifold temperature sensor
11. Left bank to right bank interconnect connector
12. Rear interstage pressure sensor
13. Exhaust gas temperature sensors.

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00600208

QSK78 (Left Side)

1. Turbocharger air inlet temperature connector
2. Rear harness box (part of the left bank harness)
3. Oil pressure sensor
4. Engine speed sensor
5. OEM 31 pin connector
6. OEM 21 pin connector
7. CENSE™ OEM 23 pin connector
8. Left bank rear intake manifold temperature sensor
9. Oil temperature sensor
10. Coolant pressure sensor
11. Timing rail pressure sensor
12. Fuel temperature sensor
13. Fuel rail pressure sensor
14. Left bank middle intake manifold temperature sensor
15. Left bank intake manifold pressure sensor

16. Quantum ECM
17. Fuel rail actuator
18. Fuel shutoff valve
19. Timing rail actuators
20. Fuses
21. CENSE™ datalink connector
22. Quantum™ datalink connector
23. Fuel pump actuator
24. Fuel pump pressure sensor
25. J1939 terminating resistor
26. Left bank front intake manifold temperature sensor
27. Rockford fan clutch solenoid
28. Coolant temperature sensor
29. Exhaust gas temperature sensors
30. Barometric pressure sensor.